

KOLMOGOROV / DESCRIPTIVE COMPLEXITY

DEF

Let x be a binary string. The minimal description of x , i.e., $d(x)$ is the shortest string $\langle M, w \rangle$ where TM M on input w halts with x on its tape.

$$K(x) = |d(x)|$$

PROPOSITION

There exists a constant c , such that for all x ,

$$K(x) \leq |x| + c \rightarrow \text{Program } M \text{ outputs input string } x$$

DEF

x is c -compressible if $K(x) \leq |x| - c$.

Otherwise, x is incompressible.

THEOREM

Incompressible strings of every length exist.

PROOF

Number of strings of length $n = 2^n$

$$\text{Number of strings of length } \leq n = \sum_{0 \leq i \leq n} 2^i = 1 + 2 + \dots + 2^{n-1} = 2^n - 1.$$

So, there exists at least one string of length n that cannot have a description of length $< n$ & hence is incompressible.

THEOREM

$K(x)$ is not computable.

PROOF

AFSOC $K(x)$ is computable.

Can construct a TM M which on input n , computes $K(x)$ for all strings x of length n & outputs the first incompressible string it finds.

Now, consider running M on a large n relative to size of M .

This results in the program M becoming a short description of the string it outputs. However, the string was incompressible by construction. Contradiction.